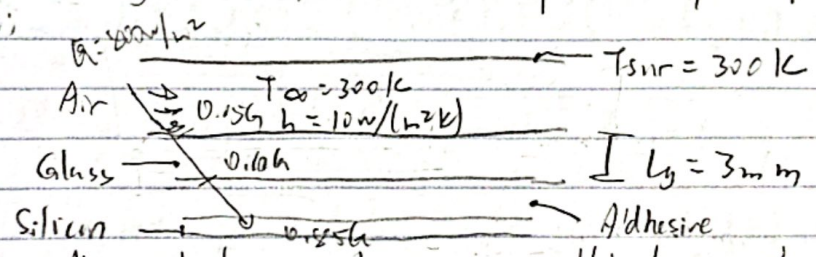
HW 4

1) Known: 3mm-thick ceria-doped glass ($k_g = 1.5$ W/m·K)
0.1mm-thick optical grade adhesive ($k_a = 1$ W/mK)
$\eta = a - bT_{si}$, $a = 0.553$, $b = 0.001\,K^{-1}$
$R_c = 10^{-4}\ m^2K/W$, $G = 800\ W/m^2$
$T_\infty = 300K$, $h = 10\ W/m^2K$
$T_{sur} = 300K$, $h_{rad} = 5\ W/m^2K$

Find: a) Thermal resistance network, value of each thermal resistance in network
b) temperatures of top surface of glass $T_{g,top}$ and silicon layer T_{si}
c) solar-to-electricity conversion efficiency η and resulting electricity power flux produced by PV panel

Schematic:



Assumptions: No contact resistance, very thin layer silicone
Backside of silicone is thermally insulated

Analysis:

a.) $T_s \xrightarrow{R''_1} T_{A,A} \xrightarrow{R''_2} T_{A,B} \xrightarrow{R''_3} T_{G,A} \xrightarrow{R''_4} T_{G,B} \xrightarrow{R''_5} T_A$, $T_{G,B} \xrightarrow{R''_6} T_A$

$R''_1 = R_c'' = \boxed{10^{-4}\ m^2K/W}$ (contact)

$R_2'' = L/k_A = 0.0001\ m/1 = \boxed{0.0001\ m^2K/W}$ (conduction)

$R_3'' = \boxed{0}$ (No contact res between glass and adhesive)

$R''_4 = L/k_g = 0.003/1.5 = \boxed{0.002\ m^2K/W}$ (conduction)

$R''_5 = 1/h = 1/10 = \boxed{0.1\ m^2K/W}$

$R''_6 = 1/h_{rad} = 1/5 = \boxed{0.2\ m^2K/W}$

R_5 and R_6 in parallel

$1/R_{tot} = 1/R''_5 + 1/R''_6$

$\boxed{R_{tot} = 0.0667\ m^2K/W}$

$\boxed{\text{Fluxes: } q''_{silicone},\ q''_{glass}}$

Assumptions: Steady state, 1D, uniform

b.) $q''_{silicone} = in - out = 0.856 - 0.856\eta$

$= .856(1-\eta) = 680(1-(0.553 - 0.001T_{si}))$

$= 680(0.447 + 0.001T_{si})$

$= 303.96 + 0.68T_{si}$

$q''_g = 0.1(800) = 80$

$q''_{tot} = 383.96 + 0.68T_{si}$

Treat rad as conv

$q''_{tot} = h(T_{g,B} - T_\infty) + h_{rad}(T_{g,B} - T_\infty)$

$383.96 + 0.68T_{si} = 10(T_{g,B} - 300) + 5(T_{g,B} - 300)$

$383.96 + 0.68T_{si} = 10T_{g,B} - 3000 + 5T_{g,B} - 1500$

$15T_{g,B} = 4883.96 + 0.68T_{si}$

$T_{g,B} = 325.6 + 0.0453T_{si}$

★ $T_{g,top} = T_{g,B}$

$q''_{tot} = \frac{T_{si} - T_{g,top}}{R} = \frac{T_{si} - T_{g,top}}{0.0001 + 0.002 + 0.0007}$

$303.96 + 0.68T_{si} = \frac{T_{si} - T_{g,top}}{0.0022}$

$T_{si} - T_{g,top} = 0.669 + 0.001496T_{si}$

plugged into algebra calc

$\boxed{T_{si} = 342.27K \quad T_{g,top} = 341.11K}$

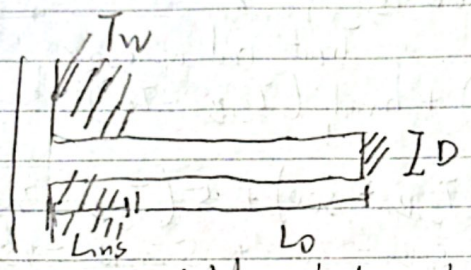
c.) $\eta = 0.553 - 0.001(342.27) = \boxed{0.211 = \eta}$

$q'' = \eta\, 0.85G = 0.211(0.85)(800) = \boxed{143.48\ W/m^2 = q''}$

2) Known: $D = 10\,mm$, $k = 80\,W/mK$, $T_w = 200°C$, $L_{ins} = 200\,mm$
$T_{max} = 100°C$ $T_\infty = 20°C$ $h = 20\,W/m^2K$

Find: a.) expression for exposed surface temp T_0
b.) calc corresponding exposed temp T_0 if $L_0 = 200\,mm$
c.) min exposed length L_0

Schematic:



Assumptions: Tip insulated, steady state

Analysis:

a.) $R_{ins} = \frac{L}{kA_c}$ $q = \frac{T_w - T_\infty}{R_{ins} + R_{fin}}$

$$R_{fin} = \frac{1}{\eta_f h A_f} = \frac{1}{\frac{\tanh mL}{mL} h A_f} = \frac{mL}{\tanh(mL)\, h A_f}$$

$$q = \frac{T_w - T_0}{R_{ins}} \quad T_0 = T_w - qR_{ins} = T_w - \frac{T_w - T_\infty}{R_{ins} + R_{fin}} R_{ins}$$

$$\boxed{T_0 = T_w - \frac{T_w - T_\infty}{L_{ins}/kA_c + \frac{mL_0}{\tanh(mL_0)\, hA_f}} \frac{L_{ins}}{kA_c}}$$

$A_c = \pi(0.005)^2 = 7.854\times10^{-5}$

$m = \sqrt{\frac{hP}{kA_c}}$ $P = 2\pi r = 0.031$

b.) $$T_0 = 200 - \left(\frac{200 - 20}{\frac{0.2}{80(7.854\times10^{-5})} + \frac{9.93(0.2)}{\tanh(9.93\cdot.2)\,20(0.006)}} \cdot \frac{0.2}{80(7.854\times10^{-5})}\right)$$

$m = \sqrt{\frac{20(0.031)}{80(7.854\times10^{-5})}} = 9.93$

$$T_0 = 200 - \left(\frac{200 - 20}{31.831 + 16.55} \cdot 31.831\right)$$

$A_{fin} = \pi(0.01)(0.2) = 0.006$

$\boxed{T_0 = 81.574°C}$ Yes, $< T_{max}$

c) $m = 9.93$

$$\frac{\theta}{\theta_b} = \frac{\cosh m(L-x)}{\cosh mL}$$

From slides

$x = L$, $\cosh 0 = 1$

$$\frac{T_0 - T_\infty}{T_w - T_\infty} = \frac{1}{\cosh mL} = \frac{100-20}{200-20} = \frac{1}{\cosh 9.93L}$$

$$0.444 = \frac{1}{\cosh 9.93L}$$

$9.93L = x$

$$0.444 = \frac{1}{\cosh x}$$

$\cosh x = 2.252$

$x = 1.452$

$1.452 = 9.93L$

$$\boxed{L = 0.146m}$$

3.) Known: $k = 400\ W/m \cdot K$, $s = 10\ mm$, $h = 100\ W/m^2K$, $\eta_f = 0.6$

Find: a) Calculate fin length L

b) thermal resistance of fin R_f and fin effectiveness ε_f

Schematic:

Assumptions: steady state

Analysis

a.) $m = \sqrt{\frac{hP}{kA_c}}$ $\quad P = 4(0.01) = 0.04$

$A_c = (0.01)^2 = 0.0001$

$m = \sqrt{\frac{100(0.04)}{400(0.0001)}} = 10\ m^{-1}$

From textbook:

$$\eta_f = \frac{\tanh mL_c}{mL_c} = \frac{\tanh(10L_c)}{10L_c} = 0.6$$

Approx: $\tanh(x) \approx x - \frac{1}{3}x^3$ $\quad x = 10L_c$

$$\frac{x - \frac{1}{3}x^3}{x} = 0.6 \qquad x \approx 1.1$$

$1.1 = 10L_c$ $\quad \boxed{L_c = 0.11\ m}$

b.) $A_f = PL$ (neglecting tip area) $= 0.04(0.11) = 0.0044\ m^2$

$$R_f = \frac{1}{\eta h A_f} = \frac{1}{0.6(100)(0.0044)} = \boxed{3.788\ K/W}$$

$$\varepsilon_f = \frac{q_f}{hA_{c,b}\theta_b} = \frac{\eta_f h A_f \theta_b}{hA_c\theta_b} \leftarrow \text{from slides} = \frac{\eta_f A_f}{A_c} = \frac{0.6(0.0044)}{0.0001}$$

$$\boxed{\varepsilon_f = 26.4}$$

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